MFE Financial Econometrics
Michaelmas 2026
Weeks 3-4 — Part I

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Course Admin

Course Admin and Housekeeping

- Make sure everyone has (e.g.) tutorial groups assigned, no timetable clashes etc
- Office hours booking system (I like Calendly)
- (Online course introduces new complications, access to material, logins, vpns, etc)
- By week 3 class reps elected?

Online-specific delivery

- ($\sim$ 15m) slides + live derivations on screen where needed
- ($\sim$ 5m) breakout rooms
- ($\sim$ 5m) report back / screen-share code/simulations/discussion
- ($\sim$ 5m) wrap up
- optional extra (short video): AR-1 conditional logL since conditional likelihood mentioned a lot

Overview

Roadmap for the next 2 weeks:

- **Estimation**
  - Maximum Likelihood (ML) method
- **Properties of Estimators**
  - Bias, Variance, Consistency, Asymptotic Normality, Efficiency
- **Distribution Theory**
  - ML, Quasi ML, Delta Method
- **Hypothesis Testing**
- **Bootstrap**
- Lecture notes: This part is covered in Chapter 2 of Prof Kevin Sheppard's lecture notes.

Estimation

Estimation

Recap: understanding the fundamentals of probability theory

- moments of distributions
- how to calculate moments, CDFs, interval probabilities
- formal definitions of (e.g.) independence

Assumed we know the underlying distribution (PDF/PMF) and parameters

$\Rightarrow$ **In reality: rarely know the distribution and/or parameters**

Estimation

BUT we observe data

- provides information about the distribution, parameters, other objects of interest
- Using the data, we should be able to learn about the distribution that generated the data, or certain moments
- $\Rightarrow$ using information contained in data, we can **estimate** unknown quantities of interest

Estimation

- **Wide range of estimation methods**
- **Frequentist**
  - We observe repeated outcomes from some data generating process (DGP), so use the data to learn about that DGP.
  - Examples:
    - Extremum estimators (M-estimators): covered in this part.
    - Linear estimators (L-estimators): covered in the next part.
- **Bayesian Estimators**
  - Make use of Bayes rule to perform inference on unknown parameters.

Key takeaway

No single method is uniformly superior to all other methods. A method can be the best option for a given setting — but in a different setting it can perform poorly.

Maximum Likelihood Method

Maximum Likelihood

- The **maximum likelihood (ML)** method is a commonly used estimation method.

- **The setting:** the econometrician knows (or is happy to assume) the underlying distribution of data, and wants to estimate parameters of that distribution.
  - coin flips follow a Bernoulli
  - stock returns: less obvious, we assume some structure

- Example: We may be happy to assume that a certain random variable has the distribution $N(\mu, \sigma^2)$ but not know the mean and variance parameters.
  - we model stock returns as Gaussian, want to learn average returns and their riskiness

What if the distribution we assume is actually wrong? The ML method is still useful

Maximum Likelihood

- Let us introduce the full setting.
- *To avoid notational complexity, we will assume that the variable of interest is continuously distributed. However, the underlying discussion applies to both discrete and continuous random variables.*

1. We observe n data points: $Y_1, \ldots, Y_n$. Define $\mathbf{Y} = (Y_1, \ldots, Y_n)$. We know the joint pdf of $\mathbf{Y}$, given by $f(\mathbf{Y}; \theta)$ where θ contains the parameters of this pdf (θ can be a vector containing multiple parameters).
2. But, we do not know the values of vector θ.

- The maximum likelihood method is based on finding the **parameter values which fit the observed data as closely as possible.**
- Technically, this is achieved by maximising the likelihood function.

Maximum Likelihood

What is the likelihood function?

- The likelihood function is simply equivalent to the pdf:

$$\mathcal{L}(\theta; \mathbf{Y}) = f(\mathbf{Y}; \theta).$$

- Note: It is standard to call this the joint likelihood function, since it is equivalent to the joint pdf of $\mathbf{Y}$.
- the maximum likelihood estimator of θ is

$$\hat{\theta} = \arg\max_{\theta} \ \mathcal{L}(\theta; \mathbf{Y}).$$

- How to make sense of this? ML is based on the idea that $\hat{\theta}$ should be the value of θ which maximises the likelihood of observing whatever dataset we have at hand.

$\Rightarrow$ We are trying to find the value of θ which makes the observed data **as likely as possible to have been observed**.

Maximum Likelihood

- When data are **independently and identically distributed** (iid), the joint likelihood function is simply the product of all marginal pdfs (last week, recall why!):

$$f(\mathbf{y};\theta) = \prod_{i=1}^{n} f(y_i;\theta).$$

- This discussion extends to **conditional pdfs**. E.g. if we want to use the distribution of $Y_1, \ldots, Y_n$ conditional on some other variable(s) X as our likelihood function:

$$f(\mathbf{y}|\mathbf{x};\theta),$$

- Note: if $Y_1, \ldots, Y_n$ are not independent unconditionally but become independent conditional on X:

$$f(\mathbf{y}|\mathbf{x};\theta) = \prod_{i=1}^{n} f(y_i|x_i;\theta).$$

- Times series context: realisations of DGP today depend on past, but the *innovations are iid*

Ex.: A Biased Coin (breakout rooms / volunteers to flip virtual coins)

Say I have a **biased coin** which flips heads with $Pr(H) = p \in (0, 1)$.

I flip the **data** sequence $\mathbf{Y} = \{H, H, T, H, T\}$, we want to estimate $\hat{p}$.

POLL: what type of coin made this run (p=?): (continuous inputs / too-hard-to-tell)

Let h_i, t_i be $0/1$ indicators, and N_H, N_T be counts:

$$\mathcal{L}(\theta; \mathbf{Y}) = \prod_{i=1}^{n} p^{h_i}(1-p)^{t_i} = p^{N_H}(1-p)^{N_T}$$

Here the algebra is easy, but not possible in all cases (We would use computational tools).

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Ex.: A Biased Coin (breakout rooms)

Warm-up (small groups of 3-5)

$\mathbf{Y} = \{H, H, T, H, T\}$

- What is the likelihood $p = 0.2$ generated this sequences? and $p = 0.7$?
- can check: $\mathcal{L}(0.7; \mathbf{Y})/\mathcal{L}(0.2; \mathbf{Y}) \approx 6$

Check the course code to make some plots. Consider and discuss :

- baseline dataset
- what about $N_H = 300, N_T = 200$?
- $N_H = 300, N_T = 20$?
- how does curvature relate to how informative the dataset is? why?
- play with the data (N_H, N_T) to get a feel for how the $\hat{p}_{ML}$ responds, report back in 5 min

Simple Computation Example: A biased Coin

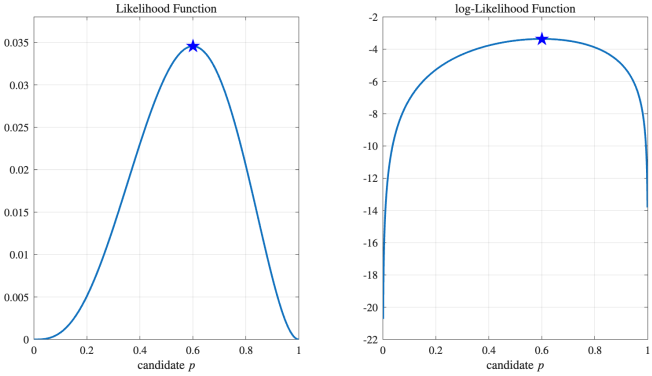

Figure 1: log-Likelihood function

Hard to discriminate between 0.4 - 0.8, quite flat! Steps to either side of 0.6 dont reduce logL much

Simple Computation Example: Few versus Many Flips

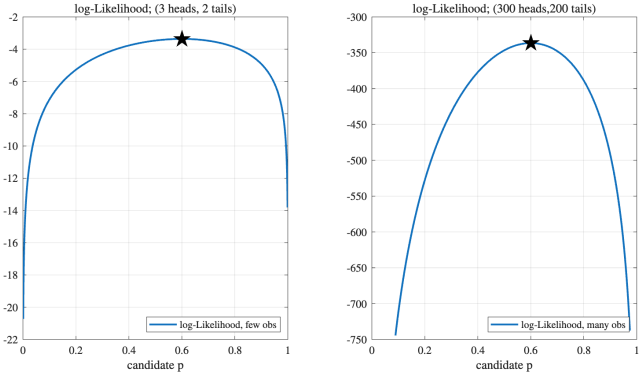

Figure 2: log-Likelihood function

More informative data: more curvature logL, easier to discriminate better/worse candidates

Maximum Likelihood

***Word of caution on notation:**

- We use $f(\mathbf{y};\theta)$ and $f(y_i;\theta)$ for the joint and marginal densities. This is a notational shorthand — the function $f(\cdot)$ need not be the same in both cases.
- A clearer notation would be for the unconditional case:

$$f(\mathbf{y};\theta) = \prod_{i=1}^{n} f_i(y_i;\theta)$$

  ... and for the conditional case:

$$f(\mathbf{y}|\mathbf{x};\theta) = \prod_{i=1}^{n} f_i(y_i|x_i;\theta)$$

- We will continue to use concise notation whenever the meaning is clear from context.

Maximum Likelihood

In practice, one usually maximises the **log-likelihood function** $\ell(\theta; \mathbf{y}) \equiv \ln \mathcal{L}(\theta; \mathbf{y})$:

$$\hat{\theta}_{ML} = \arg\max_{\theta} \ \ell(\theta; \mathbf{y}).$$

1. Since $\ln(\cdot)$ is a monotonic transformation, $\ln \mathcal{L}(\theta; \mathbf{y})$ and $\mathcal{L}(\theta; \mathbf{y})$ are maximised at the same θ.
2. The log-likelihood is usually easier to handle in proofs and in numerical optimisation.

- For some simple likelihoods (usually iid data), the ML estimator is available in **closed form**.
- When not, the maximum can be found using **numerical methods** (e.g. Newton–Raphson iteration, your favourite root-finder).

Maximum Likelihood Example: Poisson Data

Poisson data are **counts** $y_i \in \{0, 1, 2, \ldots\}$ (e.g. number of transactions over a 5-minute window).

- NB: If $Y_i \sim \text{Poisson}(\lambda)$ then $\mathbb{E}[Y_i] = \lambda$ and $\text{Var}[Y_i] = \lambda$, may or may not be desirable!

Likelihood (iid draws)[1]

$$\mathcal{L}(\lambda; \mathbf{Y}) = \prod_{i=1}^{n} \frac{\exp(-\lambda)\, \lambda^{Y_i}}{Y_i!}$$

Log-likelihood: [pause a few minutes here for students to write and solve FOC?]

$$\ell(\lambda; \mathbf{Y}) = \sum_{i=1}^{n} \left[-\lambda + Y_i \ln(\lambda) - \ln(Y_i!)\right]$$

MLE (maximise over λ, in this case FOC is easy to work with):

$$\hat{\lambda} = \frac{1}{n} \sum_{i=1}^{n} Y_i.$$

[1] useful not just for count data, e.g. $Y_i \geq 0$

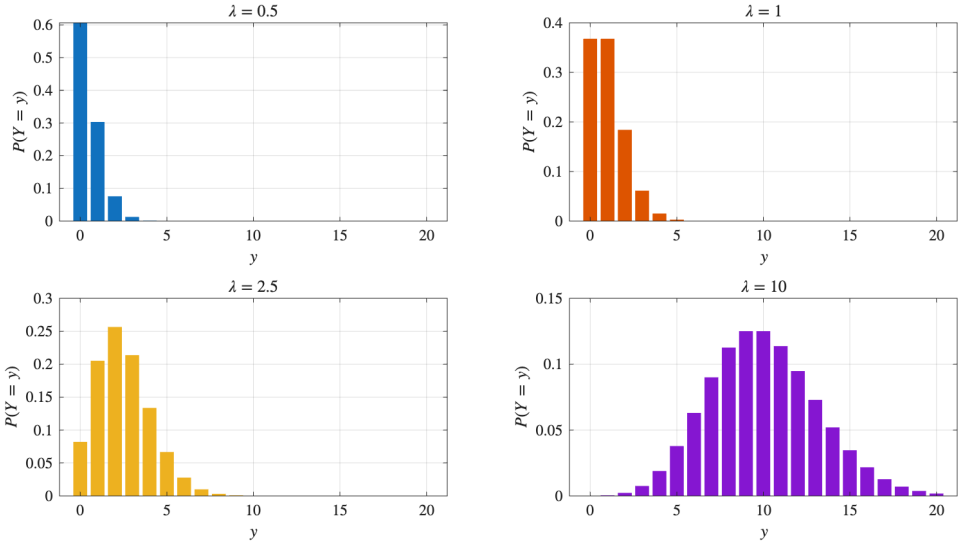

Figure 3: Poisson PMF with different λ

Conditional Maximum Likelihood

- If the model for the variable of interest is **conditional on another variable**, the likelihood function is based on the conditional probability model (plug in conditional pdfs)
- Sometimes also called **conditional maximum likelihood**.
- Maximise the conditional log-likelihood:

$$\ell(\theta; \mathbf{y}|\mathbf{x}) = \sum_{i=1}^{n} \ln f(y_i|x_i; \theta).$$

- condition on exogeneous covariates:
  - e.g.: stock returns in expansions versus recessions ($x_i \in \{0, 1\}$)
  - e.g.: oil and gas price movements normal times v crisis ($x_i \in \{0, 1\}$)
- or even condition on the history of the process itself $y_t|y_{t-1}, ..., y_1$

Optional self-study or derived on screen/short video

A correlated returns process

Consider the Gaussian Autoregressve AR(1) process for some returns data r with iid shocks:

$$r_t = \alpha + \beta r_{t-1} + \varepsilon_t; \quad \varepsilon_t \overset{iid}{\sim} N(0, \sigma^2); \quad \theta = (\alpha, \beta, \sigma^2)'$$

The density of the full history of returns can be factorised using

$$f(x, y) = f(x|y)f(y) \Rightarrow f(r_t, ..., r_1; \theta) = f(r_t|r_{t-1}, ...r_1; \theta) \times f(r_{t-1}|r_{t-2}, ...r_1; \theta)(...) \times f(r_1; \theta)$$

The conditional distribution given past observations is

$$r_t|r_{t-1}, ..., r_1 = r_t|r_{t-1} \sim N(\alpha + \beta r_{t-1}, \sigma^2)$$

The **conditional log-likelihood** simplifies to (note the starting index):

$$\ell(\theta; \mathbf{r}) = \sum_{t=2}^{T} -\frac{1}{2}\log\,(2\pi) - \frac{1}{2}\log\,\sigma^2 - \frac{1}{2}\frac{(r_t - \alpha - \beta r_{t-1})^2}{\sigma^2}$$

- condition on the first obs: $f(r_1; \theta)$ (=remove from logL) or use unconditional AR1 draw
- does this objective look familar?! it's the least squares objective!

Wrap-up: Maximum Likelihood

- The **ML method** requires the econometrician to know (or assume) the true distribution underlying the data.
- Solving the **first-order conditions** yields the ML estimator (assuming the likelihood is strictly concave).
- **Closed-form solutions** are preferable where possible — solving FOCs numerically can be time-consuming.
- **Time-varying volatility** and covariance/correlation modelling rely heavily on likelihood-based methods (next term).

Interview Questions

Teaching Approach Last Semester

MSc Finance course last semester theory heavy (model and computation, some data)

- usually multiple pathways to solution concept
- keep heterogeneity of student background in mind
- (MFE: link entrypoints to common motivation: applications in finance industry)
- visuals/graphs/plot always helpful

Emphasis on Applications of the Toolbox

- "**seeing (or better, doing) is believing**": it's one thing to read about value function convergence, another to see a code running with the result on screen doing what theory predicts
- **alternative media** (news, podcasts, the city of Glasgow itself) related to content on productive capital, useful lifetime. Theory has something to say about current AI capex boom debates
- emphasise **portable skills** (knowledge of facts and models, mechanisms, applications! coding)
- small dynamic programming **game** (recruitment assessment exercise)

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Teaching Approach, if I were to do it again...

My takeaways

- students value practical experience with problem sets / codes / tutorials a lot!
- (would require deeper restructuring of GTA hours etc)
- "alt assignment" visit some landmarks with questions on the capital (canals, docks, shipyards)
- Students were not fully able to handle incorrect AI output ("right answer" to wrong question)
  - a lot of extra $(dividend, investment \geq 0)$ constraints not in the problems
  - no interpretation or commentry at all
- $\Rightarrow$ 'debug' a bad LLM answer to build critical skills/judgement?
- $\Rightarrow$ more emphasis on "these are the features/calling cards of the correct solution"?
  - using tools in industry: has flavour of auditing model output critically

Teaching Approach, Ideas for Online MFE

For MFE students / online

- many connections b/w volatility models, uncertainty, current disruptions, asset/commodity prices
- $\Rightarrow$ (e.g.) new UK (Oxford/local) dataset of *intradaily retailer petrol prices* online soon
- online delivery:
  - real-time interactive elements (quiz/survey) or DIY asynchronous exercises
  - timezones? 2 office hours/Q+A/catch up sessions (west, east)?
  - screen-sharing can make teaching code easier?
  - more group mini-project presentations to replicate in-class student interactions?
  - (e.g.) estimate a GARCH(p,q) model on stock price of airline of your country, report to class, 5mins
- (depending on content of modules) prediction competition within family of ARMA(p,q) models?
- logistics: more office hour/1-on-1/small group slots to replace the small chitchat during breaks/after class?

Ideas for Online MFE 2

- Use of Teaching Assistants, Student Reps as "buddies"/mentors to seed group discussion sessions, slightly informal check-ins, act as facilitators, flag common confusions back to the lecturer
- Harder to see lack of understanding or engagement online v in person
- Where possible, open-source, free software for coding (matlab plug-and-play, good debugger, BUT not free).
- Video calls with a panel of past students to learn from their experience and what they are doing post degree? (makes course and pathway seem more tangible even if remote)
- **From my own studies back in the day...**
  - I still remember idiosyncratic aspects of lectures ('*cereal correlation*', '*variable instruments*)

Administration

Course

- Coordinated full assessment cycle for ECON5068: designed problem sets, code labs, exam questions, and marking rubrics; managed GTA allocation and marking consistency
- materials are portable and can scale up (towards PhD topics class) or down to a simpler Bachelor's level (more emphasis on intuition of mechanisms)
- Focus can shift more towards data/firm panel data analysis

Other

- Member of the Glasgow Research-only Staff working group
  - suggestions/actions based on cold-open/induction and networking/mentoring

Citizenship

Department

- I sit across a number of fields: try to engage with both macro and applied clusters (seminars, working groups, informal chats)
- Papers are largely collaborations with colleagues old and new (EUI, Glasgow)

Students and Juniors

- participation in PhD poster/mini conferences as discussant
- informal supervision (advice) to MSc students on dissertation topics (banking, commodity pricing, sports team valuation)
- (approach: personal interest for motivation + view towards getting a job)

Profession

- 2 recent referee reports for peer review process
- Private sector finance research conference invitation later this month (fiscal news paper)